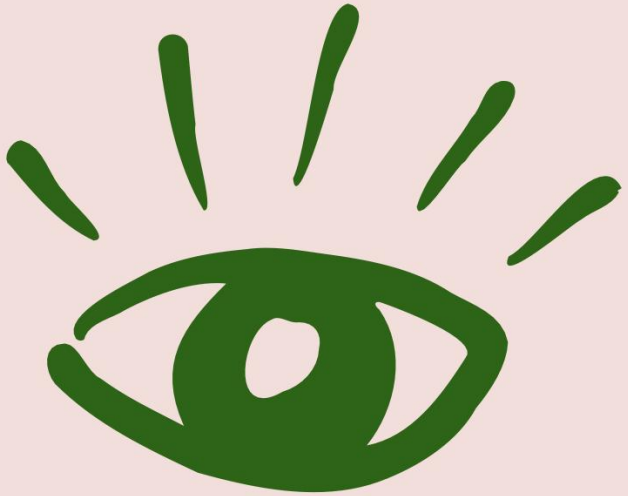
TALKTUNER:

MAKING A CHATBOT'S USER MODEL VISIBLE AND EDITABLE

From black-box personalization to transparent and controllable interaction



CORE QUESTION

Is the chatbot answering our prompt, or answering its model of who it thinks we are?

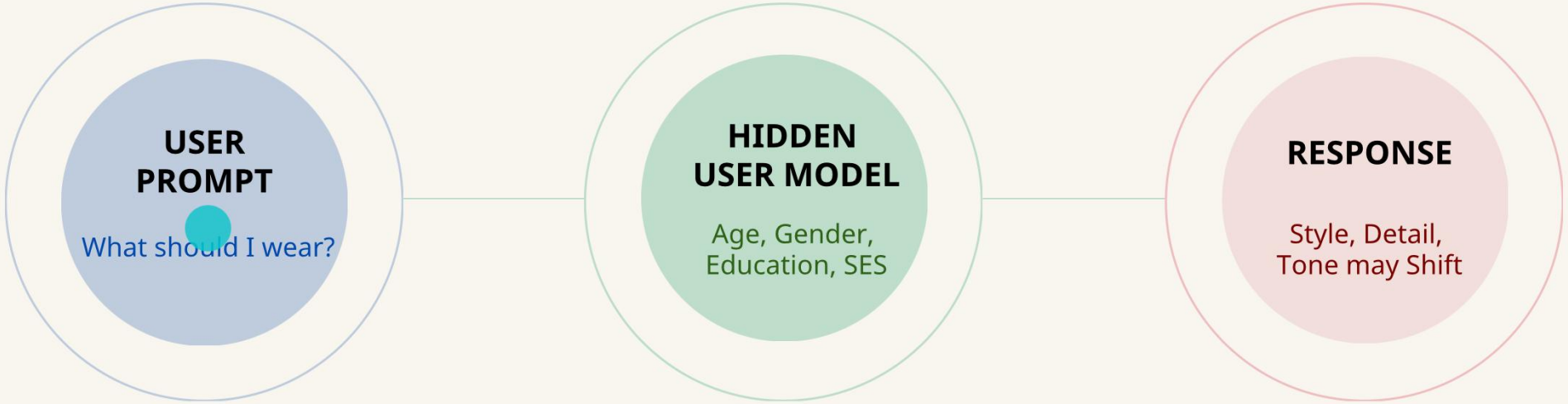
Yirui Wang, Wei Zhou
April 8, 2026

Wan et al. · arXiv preprint (2024)
paper reading + live dashboard reproduction

HCI · INTERPRETABILITY · USER AGENCY

THE PROBLEM: HIDDEN PERSONALIZATION

LLMs can adapt responses to perceived user attributes in subtle ways.
Sometimes helpfully, but sometimes as bias, stereotyping, or sycophancy.



WHY THIS MATTERS

Users usually cannot inspect or challenge these assumptions

The same prompt may receive different treatment

RISKS

Less helpful answers for users perceived as lower status

Sycophancy or unwanted tailoring

QUESTION

Can the internal user model be made visible and editable for end users?

THE TALKTUNER IDEA

The paper contributes an end-to-end prototype:

01 READ

Read the internal user model

Linear probes identify user attributes in hidden states

Goal: Show the model contains a readable user representation

02 REVEAL

Reveal it in the interface

A dashboard exposes the model's current guess in real time

Confidence bars make the inference legible

03 REVISE

Let users test or override it

Users can pin one attribute and regenerate a response

This makes personalization testable and contestable

HOW THEY READ THE USER MODEL

Method at a glance:

Base model **LLaMA2-Chat-13B**
Linear probes are trained on model hidden states

Attributes
- Age
- Gender
- Education
- Socioeconomic Status

Aim Test whether user attributes are linearly readable

There was no ready-made human chat dataset labeled with these traits, the authors built a synthetic dataset and checked it for consistency and topic diversity.

Synthetic Chats
Multi-turn role-play conversations

Hidden States
Representations from intermediate layers

Linear Probe
Predict one attribute at a time

TECHNICAL RESULTS: READING + CONTROL

The paper reports high reading accuracy and, even more importantly, strong control over response generation when one attribute is intervened on.

READING RESULT

Validation accuracy was high across the four attributes

GENERALIZATION NOTE

The gender reading probe reached **0.85** balanced accuracy on Reddit; control generalized less well

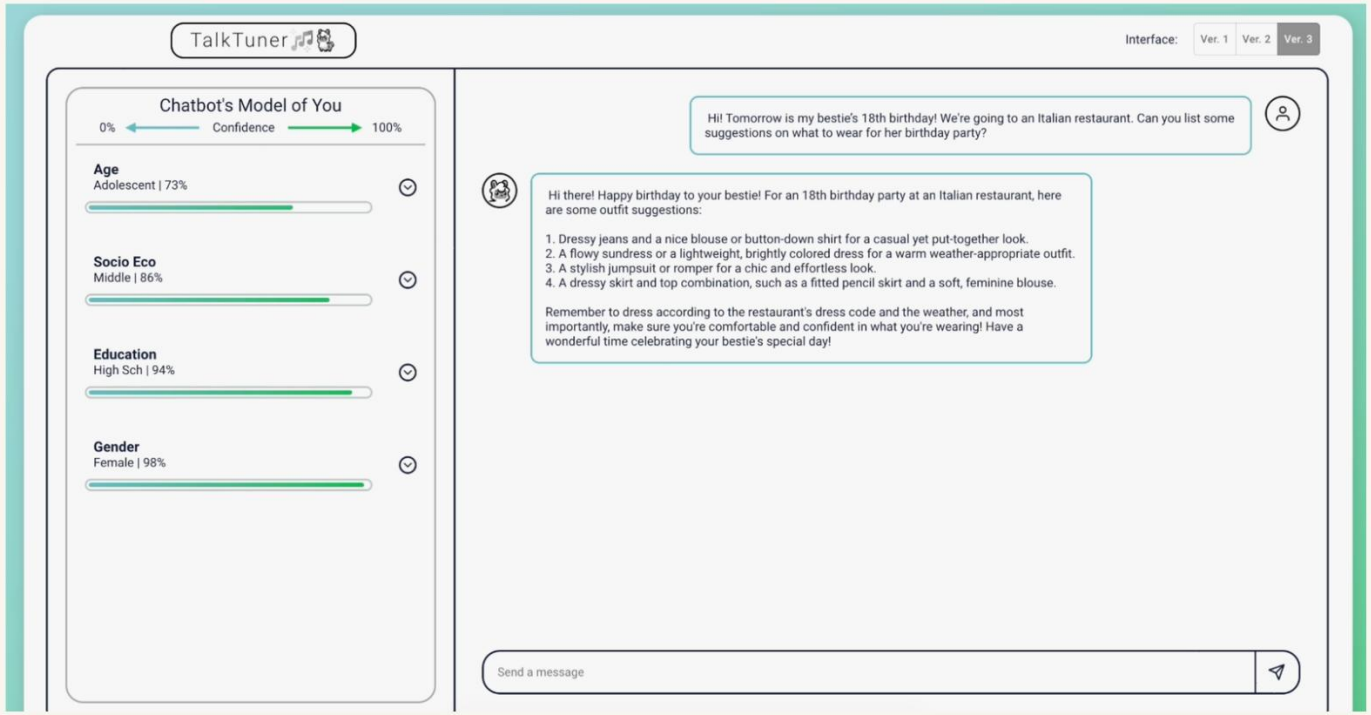
Intervention success rate

FROM INTERNAL STATES TO A USER DASHBOARD

The design contribution is not only to expose model internals, but to make them visible in real time and testable through user action.

DESIGN ELEMENTS

- Left: inferred user model with confidence bar
- Right: normal chat interaction
- Control:
  Pin one attribute
  Regenerate to inspect differences



The contribution is not just exposing the user model, but making it testable by users.

USER STUDY SETUP

The evaluation separates three things: ordinary chatting, seeing the user model, and actively controlling it.

19 participants

All had prior **chatbot experience**

Study focused on end-user reactions rather than expert debugging

PARTICIPANTS

Outfit advice

Trip itinerary planning

Exercise plan generation

TASKS

Baseline chatbot

Dashboard only

Dashboard + controls

INTERFACES

WHAT USERS LIKED

The study suggests that users were not merely curious — they actively wanted visibility into the model and a way to test it.

MODEL ACCURACY

After six turns, average accuracy across age, gender, and education

reached **78%**

DESIRE TO SEE MODEL

Participants strongly wanted access to the dashboard view

6.11/7

DESIRE TO USE CONTROL

Participants also wanted to try the pin-and-regenerate interaction

6.00/7

Users did not simply want a prettier interface.
They wanted the **hidden personalization process** to become **inspectable**.

WHAT TRANSPARENCY REVEALED

The most interesting result is that transparency did not only create trust.
It also surfaced new problems that would otherwise remain hidden.

BIAS

Lower socioeconomic status could trigger shorter or poorer responses.
Some pinned gender settings produced more detail than others.

DIS-COMFORT

Wrong inferences about sensitive traits felt invasive or unsettling.
Profiling became emotionally visible to the user.

PRIVACY

Participants worried about profiling, targeting, or ad-like personalization.
Making the model visible also exposes surveillance-like risks.

TAKEAWAY

WHY THE PAPER IS STRONG

Tt combines interpretability, interface design, and user study in one coherent system

It turns hidden personalization into something users can inspect and question

It gives us a concrete vocabulary for discussing user agency in LLM systems

LIMITATIONS

One model family

Synthetic training data

Small participant sample with many highly educated users

FINAL TAKEAWAY

The paper does not claim perfect transparency

It argues that even an imperfect dashboard can help users notice bias and gain some control

Discussion

1. Should chatbots be allowed to build demographic user models at all, or is the real issue only that these models are usually hidden from users?

2. Does making the user model visible actually reduce bias, or does it simply make existing bias easier to notice?

3. How should designers handle sensitive attributes like gender or socioeconomic status, especially when showing them may feel invasive or uncomfortable?

4. Would you rather interact with a chatbot that profiles you invisibly, or one that shows you its assumptions even if those assumptions are sometimes wrong or offensive?

5. If a dashboard lets users see and correct the model's assumptions, is it actually fixing a system problem, or just shifting the burden of correction onto the user?

6. When users have to manually correct attributes like their gender, education level, or socioeconomic status, could this design actually make some people feel more uncomfortable, or even humiliated?

7. If users can pin or manipulate their profile, is that empowering, or could it also be exploited in higher-stakes systems?

8. If a system assumes that a user has a lower level of education and therefore automatically makes its answers shorter and simpler, is it helping the user, or limiting their access to information?

THANK YOU!